

# Paul Prae

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## PROFESSIONAL SUMMARY

AI engineering leader with 15 years of experience architecting enterprise AI, ML, and data solutions across healthcare, life science, and insurance. Proven track record at Amazon Web Services, Microsoft, Booz Allen Hamilton, and Slalom Consulting delivering production AI systems for Fortune 500 clients — from clinical decision support platforms serving 50M+ health plan members to federated research platforms advancing brain health. Combines full-stack technical depth across data engineering, ML systems, and cloud infrastructure with demonstrated leadership in building cross-functional teams, coaching executives, and driving AI strategy. Brings deep healthcare domain expertise and responsible AI governance practices to drive transformative outcomes at the intersection of AI and regulated industries.

## PROFESSIONAL EXPERIENCE

### Staff AI DataOps Engineer

Arine | Remote | Mar 2025 – Present

- Designed the AI DataOps Toolkit — an agentic development framework with governance standards, reference architectures, and change management procedures — and led AI enablement workshops training groups of up to 100 engineers on AI-first development practices.
- Developed HIPAA-compliant coding assistants including a Data Engineering Agent that autonomously generated, deployed, and tested ETL pipelines and a Dev Environment Setup Agent that eliminated multi-day onboarding friction.
- Built an observability pipeline monitoring Arine's autonomous pharmacist AI agent — ingesting production behavior data from DynamoDB into Snowflake, transforming with dbt, and visualizing performance metrics in QuickSight dashboards.
- Managed enterprise data platform on Snowflake and AWS processing petabytes of healthcare data from hundreds of heterogeneous sources, supporting AI/ML systems serving 45+ health plans and 50M+ members.
- Architected high-throughput CDC pipelines from PostgreSQL to Snowflake using serverless, event-driven architectures (Lambda, Step Functions, EventBridge, SNS/SQS).

### Chief AI Architect, Senior Manager

Booz Allen Hamilton | Atlanta, GA | Jul 2024 – Mar 2025

- Architected AI and analytics solutions for healthcare and life science clients at a Fortune 500 consulting firm, enabling population health insights from clinical, behavioral, and genomic data across FedRAMP and HIPAA-compliant environments.
- Led cross-functional teams of engineers, data scientists, and domain experts to design and deploy AI software products, addressing algorithmic bias and privacy risks in regulated clinical settings.
- Advised client leadership on AI operationalization strategy, driving adoption of data-driven medicine across rapidly evolving clinical, regulatory, and technological landscapes.

### Neuroinformatics Data Platform Architect

TRENDS Center | Atlanta, GA | Jan 2022 – Sep 2023

- Led architecture of COINSTAC, an open-source federated analysis platform for neuroimaging and genomics research, enabling decentralized collaboration with differential privacy protections across research institutions.
- Built GPU-focused compute infrastructure on AWS for deep learning workloads; drove platform commercialization through SBIR grant writing, open-source contributor recruitment, and co-authored research published in Frontiers in Neuroinformatics.

### Chief AI Officer, Founder

Hyperbloom | United States | Jan 2020 – Sep 2025

- Founded and scaled an AI consulting practice to \$1.4M ARR from a \$40K investment, delivering cloud-based AI and data solutions to healthcare, life science, and financial services clients.
- Architected enterprise solutions including disaster recovery for clinical trial software supporting 10,000+ research sites across 45 countries (5.5M monthly monitoring activities) and blockchain-based data platforms meeting HIPAA, GDPR, and CCPA compliance with HL7 interoperability.

### Enterprise AI and ML Solutions Architect

Amazon Web Services | Atlanta, GA | Aug 2018 – May 2021

- Managed a portfolio of 10+ enterprise accounts — including Cox Enterprises, Cox Automotive, Equifax, NCR, Accenture, and Deloitte — delivering AI/ML architecture guidance, proof-of-concept development, and benchmarking to drive production adoption.

- Engaged nationally as a subject matter expert in AI and ML, authoring technical white papers and reference architectures and speaking at private and public AWS events.
- Partnered with C-suite executives on technical and business strategy — including go-to-market, cost management, and technical recruiting — driving AWS AI/ML adoption and revenue.

### **Advanced Analytics Consultant**

**Slalom Consulting** | Atlanta, GA | Jul 2015 – Jan 2018

- Delivered 7+ client engagements across healthcare and public sector, including a behavioral healthcare application generating \$2M+ in revenue and statewide demand forecasting of crisis services across dozens of Georgia facilities.
- Recognized nationally as a subject matter expert in NLP and ML; grew the AI practice through workshops, technical presentations, and sales support across the Microsoft Azure and Cortana Intelligence ecosystems.

### **Software Engineer**

**Red Ventures** | Fort Mill, SC | Mar 2014 – Jul 2015

- Engineered full-stack marketing platform serving 16+ Fortune 500 clients (DirecTV, Verizon, AT&T, Progressive, Liberty Mutual), developing proprietary CMS, experience management platform, and hundreds of high-traffic consumer-facing websites.

### **Senior Support Engineer**

**Microsoft** | Charlotte, NC | Jul 2012 – Mar 2014

- Delivered advanced technical support for Fortune 100 clients (Chevron, Walmart, Disney, Bank of America, Intel, BlueCross BlueShield) across hundreds of high-impact web applications and server farms; nominated for Global Business Support's "Entrepreneur of the Year" (2013).

## **EDUCATION**

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### **Bachelor of Science, Computer Science (AI Emphasis) & Bachelor of**

Arts, Cognitive Science (AI Focus) **University of Georgia** | 2008 – 2012 Graduated Cum Laude. Dean's List. HOPE Scholarship recipient.

## **TECHNICAL SKILLS**

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**AI & Machine Learning:** Generative AI, Large Language Models, Multi-Agent Systems, NLP, Deep Learning, MLOps, Prompt Engineering, Predictive Analytics, Conversational AI

**Cloud & Infrastructure:** AWS (Bedrock, SageMaker, ECS/EKS, Lambda, S3, DynamoDB, Glue, QuickSight, Step Functions), Microsoft Azure, Snowflake, Docker, Distributed Systems

**Data Engineering:** dbt, PostgreSQL, Apache Kafka, Neo4j, ETL/CDC Pipelines, Data Warehousing, Data Governance, HIPAA/GDPR Compliance

**Programming Languages:** Python, SQL, JavaScript/TypeScript, C#, Bash, Java, PowerShell

**AI Tools & Frameworks:** Cursor AI, Claude Code, GitHub Copilot, LangChain, Ollama, n8n, Open WebUI, Amazon Bedrock, AWS SageMaker

**Leadership & Strategy:** Engineering Team Management, Solutions Architecture, AI Governance, Executive Coaching (93% client performance improvement), Agile/Scrum, Technical Pre-Sales

## **CERTIFICATIONS**

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- **AWS Certified Machine Learning Engineer** — Amazon Web Services (Sep 2025)
- **AWS Certified AI Practitioner** — Amazon Web Services (Sep 2025)
- **AWS Certified Machine Learning - Specialty** — Amazon Web Services (Jan 2020)
- **AWS Certified Solutions Architect** — Amazon Web Services (Jul 2018)

## **PROJECTS**

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### **Modular Earth**

Founded a not-for-profit building open-source AI agents to help working-class families generate wealth. Active projects include a career coaching assistant and financial planning assistant built on a privacy-preserving, local-first stack (Ollama, LangChain, n8n, Neo4j). Vision: deploy AI-as-a-public-service in local libraries at no cost.

## **AI Engineering Assistant**

Open-source prompt engineering and context engineering system that helps architects design agentic workflows for autonomous task execution. Integrates with Cursor and GitHub Copilot, providing step-by-step guidance, built-in validation, and self-improving prompt optimization.

## **PUBLICATIONS**

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[Enhancing Collaborative Neuroimaging Research: COINSTAC Vaults for Federated Analysis and Reproducibility](#) Frontiers in Neuroinformatics (Jun 2023). Co-authored peer-reviewed research on federated neuroimaging analysis, enabling multi-site research without sharing raw data while improving reproducibility and increasing sample sizes across institutions.